

Отчёт о лабораторной работе

Лабораторная работа 15

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1 Цель работы

Получение навыков по работе с журналами системных событий

2 Выполнение лабораторной работы

На сервере перейдем в каталог конфигурации rsyslog и создадим файл netlog-server.conf для настройки централизованного сбора логов (рис. 2.1).

```
[nsandryushin@server.nsandryushin.net ~]$ cd /etc/rsyslog.d
[nsandryushin@server.nsandryushin.net rsyslog.d]$ sudo touch netlog-server.conf
[sudo] password for nsandryushin:
[nsandryushin@server.nsandryushin.net rsyslog.d]$ sudo nano netlog-server.conf
```

Рис. 2.1: Создание файла конфигурации на сервере

В открывшемся редакторе внесем настройки для включения модуля imtcp и прослушивания порта 514 по протоколу TCP (рис. 2.2).

```
GNU nano 8.1 netlog-server.conf
$ModLoad imtcp
$InputTCPServerRun 514
```

Рис. 2.2: Редактирование конфигурации rsyslog на сервере

Перезапустим службу rsyslog для применения изменений и начнем проверку прослушиваемых портов с помощью утилиты lsof, отфильтровав вывод по TCP (рис. 2.3).

```
[nsandryushin@server.nsandryushin.net rsyslog.d]$ sudo systemctl restart rsyslog
[nsandryushin@server.nsandryushin.net rsyslog.d]$ sudo lsof | grep TCP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1001/gvfs
Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1001/doc
Output information may be incomplete.
```

systemd	1	root	42u	IPv4	6235	0t0	TCP	*:sun
rpc (LISTEN)								
systemd	1	root	44u	IPv6	6247	0t0	TCP	*:sun
rpc (LISTEN)								
systemd	1	root	91u	IPv6	9058	0t0	TCP	*:web
sm (LISTEN)								
rpcbind	861	rpc	5u	IPv4	6235	0t0	TCP	*:sun
rpc (LISTEN)								
rpcbind	861	rpc	7u	IPv6	6247	0t0	TCP	*:sun
rpc (LISTEN)								
cupsd	1264	root	7u	IPv6	11459	0t0	TCP	local
host:ipp (LISTEN)								
cupsd	1264	root	8u	IPv4	11460	0t0	TCP	local
host:ipp (LISTEN)								
sshd	1274	root	7u	IPv4	9903	0t0	TCP	*:dow
n (LISTEN)								
sshd	1274	root	8u	IPv6	9905	0t0	TCP	*:dow
n (LISTEN)								
sshd	1274	root	9u	IPv4	9907	0t0	TCP	*:ssh
(LISTEN)								
sshd	1274	root	10u	IPv6	9909	0t0	TCP	*:ssh
(LISTEN)								
named	1377	named	27u	IPv4	11674	0t0	TCP	local
host:domain (LISTEN)								
named	1377	named	28u	IPv4	11675	0t0	TCP	local

Рис. 2.3: Перезапуск службы и начало проверки портов

В продолжении вывода команды `lsof` убедимся, что процессы `rsyslogd` успешно прослушивают порт 514 на всех интерфейсах (рис. 2.4).

```

rsyslogd 8324 root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 8327 in:imjour root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 8327 in:imjour root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 8328 in:imtcp root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 8328 in:imtcp root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 8329 in:imtcp root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 8329 in:imtcp root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 8330 in:imtcp root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 8330 in:imtcp root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 8331 in:imtcp root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 8331 in:imtcp root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 8332 in:imtcp root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 8332 in:imtcp root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 8335 rs:main root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN)
rsyslogd 8324 8335 rs:main root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN)
[nsandryushin@server.nsandryushin.net rsyslog.d]$ █

```

Рис. 2.4: Подтверждение прослушивания порта 514

Настроим межсетевой экран на сервере, разрешив прием соединений по порту 514/tcp, и сохраним правило как постоянное (рис. 2.5).

```

[nsandryushin@server.nsandryushin.net rsyslog.d]$ sudo firewall-cmd --add-port=514/tcp
success
[nsandryushin@server.nsandryushin.net rsyslog.d]$ sudo firewall-cmd --add-port=514/tcp --permanent
success
[nsandryushin@server.nsandryushin.net rsyslog.d]$ █

```

Рис. 2.5: Настройка firewall на сервере

Перейдем на клиентскую машину. В каталоге /etc/rsyslog.d создадим файл конфигурации netlog-client.conf (рис. 2.6).

```
[nsandryushin@client.nsandryushin.net ~]$ cd /etc/rsyslog.d
[nsandryushin@client.nsandryushin.net rsyslog.d]$ sudo netlog-client.conf
[sudo] password for nsandryushin:
sudo: netlog-client.conf: command not found
[nsandryushin@client.nsandryushin.net rsyslog.d]$ sudo touch netlog-client.conf
[nsandryushin@client.nsandryushin.net rsyslog.d]$ nano netlog-client.conf
```

Рис. 2.6: Создание файла конфигурации на клиенте

В файле конфигурации пропишем правило отправки всех журналов (.) на сервер по адресу `server.nsandryushin.net` через порт 514 по протоколу TCP (рис. 2.7).

```
GNU nano 8.1 netlog-client.conf
*. * @@server.nsandryushin.net:514
```

Рис. 2.7: Редактирование конфигурации rsyslog на клиенте

Вернемся на сервер и запустим отслеживание файла `/var/log/messages` в реальном времени, чтобы убедиться в поступлении логов (рис. 2.8).

```

[nsandryushin@server.nsandryushin.net rsyslog.d]$ sudo tail -f /var/log/messages
Dec 13 18:31:39 server systemd[1]: Stopped user-runtime-dir@0.service - User Runtime Directory /run/user/0.
Dec 13 18:31:39 server systemd[1]: Removed slice user-0.slice - User Slice of UID 0.
Dec 13 18:33:20 server systemd-logind[924]: Existing logind session ID 3 used by new audit session, ignoring.
Dec 13 18:33:20 server systemd[1]: Created slice user-0.slice - User Slice of UID 0.
Dec 13 18:33:20 server systemd[1]: Starting user-runtime-dir@0.service - User Runtime Directory /run/user/0...
Dec 13 18:33:20 server systemd-logind[924]: New session c8 of user root.
Dec 13 18:33:20 server systemd[1]: Finished user-runtime-dir@0.service - User Runtime Directory /run/user/0.
Dec 13 18:33:20 server systemd[1]: Starting user@0.service - User Manager for UID 0...
Dec 13 18:33:20 server systemd-logind[924]: New session 7 of user root.
Dec 13 18:33:20 server systemd[8447]: Queued start job for default target default.target.
Dec 13 18:33:20 server systemd[8447]: Created slice app.slice - User Application Slice.
Dec 13 18:33:20 server systemd[8447]: grub-boot-success.timer - Mark boot as successful after the user session has run 2 minutes was skipped because of an unmet condition check (ConditionUser=!@system).
Dec 13 18:33:20 server systemd[8447]: Started systemd-tmpfiles-clean.timer - Daily Cleanup of User's Temporary Directories.
Dec 13 18:33:20 server systemd[8447]: Reached target paths.target - Paths.
Dec 13 18:33:20 server systemd[8447]: Reached target timers.target - Timers.
Dec 13 18:33:20 server systemd[8447]: Starting dbus.socket - D-Bus User Message Bus Socket...
Dec 13 18:33:20 server systemd[8447]: pipewire-pulse.socket - PipeWire PulseAudio was skipped because of an unmet condition check (ConditionUser=!root).
Dec 13 18:33:20 server systemd[8447]: Listening on pipewire.socket - PipeWire Multimedia System Sockets.
Dec 13 18:33:20 server systemd[8447]: Starting systemd-tmpfiles-setup.service - Create User Files and Directories...
Dec 13 18:33:20 server systemd[8447]: Finished systemd-tmpfiles-setup.service - Create User Files and Directories.
Dec 13 18:33:20 server systemd[8447]: Listening on dbus.socket - D-Bus User Message Bus Socket.
Dec 13 18:33:20 server systemd[8447]: Reached target sockets.target - Sockets.
Dec 13 18:33:20 server systemd[8447]: Reached target basic.target - Basic System.
Dec 13 18:33:20 server systemd[8447]: Reached target default.target - Main User Target.
Dec 13 18:33:20 server systemd[8447]: Startup finished in 259ms.
Dec 13 18:33:20 server systemd[1]: Started user@0.service - User Manager for UID 0.
Dec 13 18:33:20 server systemd[1]: Started session-c8.scope - Session c8 of User root.

```

Рис. 2.8: Просмотр логов на сервере через tail

Запустим графическую утилиту мониторинга системы. В открывшемся окне gnome-system-monitor просмотрим список активных процессов и потребляемые ими ресурсы (рис. 2.9).

```
[nsandryushin@server.nsandryushin.net rsyslog.d]$ gnome-system-monitor
```

Process Name	User	% CPU	ID	Memory	Disk read total	Disk write total	Disk read	Disk write	Priority
at-spi2-registryd	nsandryushin	0.00	6416	131.1 kB	708.6 kB	N/A	N/A	N/A	Normal
at-spi-bus-launcher	nsandryushin	0.00	6408	262.1 kB	667.6 kB	N/A	N/A	N/A	Normal
bash	nsandryushin	0.00	7887	1.6 MB	9.0 MB	N/A	N/A	N/A	Normal
bash	nsandryushin	0.00	8477	2.0 MB	843.8 kB	N/A	N/A	N/A	Normal
catatonit	nsandryushin	0.00	7843	N/A	663.6 kB	N/A	N/A	N/A	Normal
dbus-broker	nsandryushin	0.00	6287	1.3 MB	2.1 MB	N/A	N/A	N/A	Normal
dbus-broker	nsandryushin	0.00	6415	131.1 kB	184.3 kB	N/A	N/A	N/A	Normal
dbus-broker-launch	nsandryushin	0.00	6286	N/A	254.0 kB	N/A	N/A	N/A	Normal
dbus-broker-launch	nsandryushin	0.00	6414	N/A	N/A	N/A	N/A	N/A	Normal
dconf-service	nsandryushin	0.00	6440	393.2 kB	479.2 kB	614.4 kB	N/A	N/A	Normal
evolution-addressbook-factory	nsandryushin	0.00	6702	131.1 kB	3.8 MB	53.2 kB	N/A	N/A	Normal
evolution-alarm-notify	nsandryushin	0.00	6504	262.1 kB	4.5 MB	N/A	N/A	N/A	Normal
evolution-calendar-factory	nsandryushin	0.00	6622	917.5 kB	4.3 MB	N/A	N/A	N/A	Normal
evolution-source-registry	nsandryushin	0.00	6454	131.1 kB	3.0 MB	N/A	N/A	N/A	Normal
firefox	nsandryushin	4.54	7118	156.3 MB	469.3 MB	49.6 MB	2.7 KiB/s	N/A	Normal
gdm-wayland-session	nsandryushin	0.00	6282	N/A	20.5 kB	N/A	N/A	N/A	Normal
gjs	nsandryushin	0.00	6460	32.8 kB	2.3 MB	N/A	N/A	N/A	Normal
gjs	nsandryushin	0.00	6624	3.3 MB	5.2 MB	N/A	N/A	N/A	Normal
gnome-keyring-daemon	nsandryushin	0.00	6269	323.6 kB	2.8 MB	4.1 kB	N/A	N/A	Normal
gnome-session-binary	nsandryushin	0.00	6289	131.1 kB	520.2 kB	N/A	N/A	N/A	Normal
gnome-session-binary	nsandryushin	0.00	6342	655.4 kB	3.2 MB	4.1 kB	N/A	N/A	Normal
gnome-session-ctl	nsandryushin	0.00	6340	N/A	24.6 kB	N/A	N/A	N/A	Normal
gnome-shell	nsandryushin	22.22	6369	202.1 MB	322.3 MB	139.3 kB	1.3 KiB/s	N/A	Normal
gnome-shell-calendar-server	nsandryushin	0.00	6432	786.4 kB	5.4 MB	N/A	N/A	N/A	Normal
gnome-software	nsandryushin	0.00	6528	6.8 MB	68.6 MB	N/A	N/A	N/A	Normal
gnome-system-monitor	nsandryushin	5.95	8506	144.9 MB	48.6 MB	53.2 kB	N/A	N/A	Normal
goa-daemon	nsandryushin	0.00	6587	131.1 kB	659.5 kB	N/A	N/A	N/A	Normal
goa-identity-service	nsandryushin	0.00	6691	131.1 kB	397.3 kB	N/A	N/A	N/A	Normal
gsd-ally-settings	nsandryushin	0.00	6464	131.1 kB	233.5 kB	N/A	N/A	N/A	Normal
gsd-color	nsandryushin	0.00	6466	262.1 kB	970.8 kB	N/A	N/A	N/A	Normal
gsd-datetime	nsandryushin	0.00	6475	131.1 kB	307.2 kB	N/A	N/A	N/A	Normal
gsd-disk-utility-notify	nsandryushin	0.00	6505	131.1 kB	159.7 kB	N/A	N/A	N/A	Normal
gsd-housekeeping	nsandryushin	0.00	6483	393.2 kB	647.2 kB	N/A	N/A	N/A	Normal
gsd-keyboard	nsandryushin	0.00	6484	262.1 kB	1.8 MB	N/A	N/A	N/A	Normal
gsd-media-keys	nsandryushin	0.00	6485	917.5 kB	1.8 MB	N/A	N/A	N/A	Normal

Рис. 2.9: Интерфейс Gnome System Monitor

Установим продвинутый просмотрщик логов lnav на сервере, скопировав исполняемый файл в системную директорию /usr/bin (рис. 2.10).

```
[nsandryushin@server.nsandryushin.net rsyslog.d]$ cd ~/Downloads/
[nsandryushin@server.nsandryushin.net Downloads]$ sudo cp lnav /usr/bin
[nsandryushin@server.nsandryushin.net Downloads]$
```

Рис. 2.10: Ручная установка lnav на сервере

Запустим lnav на сервере и посмотрим журналы в удобном структурированном виде с подсветкой синтаксиса (рис. 2.11).

```
2025-12-13T18:38:26 UTC Press F2 to enable mouse support
LOG v : 2025-12-13T18:38:21.000 : syslog_log : messages[6050] : systemd[1] :
Dec 13 18:38:18 server systemd[8447]: Stopped systemd-tmpfiles-clean.timer - Daily Cleanup of User's Te
Dec 13 18:38:18 server systemd[8447]: Closed dbus.socket - D-Bus User Message Bus Socket.
Dec 13 18:38:18 server systemd[8447]: Closed pipewire.socket - PipeWire Multimedia System Sockets.
Dec 13 18:38:18 server systemd[8447]: Stopped systemd-tmpfiles-setup.service - Create User Files and Di
Dec 13 18:38:18 server systemd[8447]: Removed slice app.slice - User Application Slice.
Dec 13 18:38:18 server systemd[8447]: Reached target shutdown.target - Shutdown.
Dec 13 18:38:18 server systemd[8447]: Finished systemd-exit.service - Exit the Session.
Dec 13 18:38:18 server systemd[8447]: Reached target exit.target - Exit the Session.
Dec 13 18:38:18 server systemd-logind[924]: Removed session 7.
Dec 13 18:38:18 server systemd[1]: user@0.service: Deactivated successfully.
Dec 13 18:38:18 server systemd[1]: Stopped user@0.service - User Manager for UID 0.
Dec 13 18:38:18 server systemd[1]: Stopping user-runtime-dir@0.service - User Runtime Directory /run/us
Dec 13 18:38:18 server systemd[1]: run-user-0.mount: Deactivated successfully.
Dec 13 18:38:18 server systemd[1]: user-runtime-dir@0.service: Deactivated successfully.
Dec 13 18:38:18 server systemd[1]: Stopped user-runtime-dir@0.service - User Runtime Directory /run/use
Dec 13 18:38:18 server systemd[1]: Removed slice user-0.slice - User Slice of UID 0.
Dec 13 18:38:18 server systemd[1]: user-0.slice: Consumed 1.465s CPU time, 204.3M memory peak, 3.8M mem
Dec 13 18:38:20 server systemd-logind[924]: Existing logind session ID 3 used by new audit session, ign
Dec 13 18:38:20 server systemd[1]: Created slice user-0.slice - User Slice of UID 0.
Dec 13 18:38:20 server systemd[1]: Starting user-runtime-dir@0.service - User Runtime Directory /run/us
Dec 13 18:38:20 server systemd-logind[924]: New session c11 of user root.
Dec 13 18:38:20 server systemd[1]: Finished user-runtime-dir@0.service - User Runtime Directory /run/us
Dec 13 18:38:20 server systemd[1]: Starting user@0.service - User Manager for UID 0...
Dec 13 18:38:20 server systemd-logind[924]: New session 8 of user root.
Dec 13 18:38:21 server systemd[10496]: Queued start job for default target default.target.
Dec 13 18:38:21 server systemd[10496]: Created slice app.slice - User Application Slice.
Dec 13 18:38:21 server systemd[10496]: grub-boot-success.timer - Mark boot as successful after the user
Dec 13 18:38:21 server systemd[10496]: Started systemd-tmpfiles-clean.timer - Daily Cleanup of User's T
Dec 13 18:38:21 server systemd[10496]: Reached target paths.target - Paths.
Dec 13 18:38:21 server systemd[10496]: Reached target timers.target - Timers.
Dec 13 18:38:21 server systemd[10496]: Starting dbus.socket - D-Bus User Message Bus Socket...
Dec 13 18:38:21 server systemd[10496]: pipewire-pulse.socket - PipeWire PulseAudio was skipped because
Dec 13 18:38:21 server systemd[10496]: Listening on pipewire.socket - PipeWire Multimedia System Socket
Dec 13 18:38:21 server systemd[10496]: Starting systemd-tmpfiles-setup.service - Create User Files and
Dec 13 18:38:21 server systemd[10496]: Listening on dbus.socket - D-Bus User Message Bus Socket.
Dec 13 18:38:21 server systemd[10496]: Finished systemd-tmpfiles-setup.service - Create User Files and
Dec 13 18:38:21 server systemd[10496]: Reached target sockets.target - Sockets.
Dec 13 18:38:21 server systemd[10496]: Reached target basic.target - Basic System.
Dec 13 18:38:21 server systemd[10496]: Reached target default.target - Main User Target.
Dec 13 18:38:21 server systemd[10496]: Startup finished in 196ms.
Dec 13 18:38:21 server systemd[1]: Started user@0.service - User Manager for UID 0.
Dec 13 18:38:21 server systemd[1]: Started session-c11.scope - Session c11 of User root.
```

Рис. 2.11: Просмотр логов через lnav на сервере

Аналогичным образом скопируем утилиту lnav на клиентскую машину и запустим её (рис. 2.12).

```
[nsandryushin@client.nsandryushin.net client]$ cd ~/Downloads/
[nsandryushin@client.nsandryushin.net Downloads]$ ls
015-netlog.pdf lnav lnav-0.13.2-linux-musl-x86_64 lnav-0.13.2-linux-musl-x86_64.zip
[nsandryushin@client.nsandryushin.net Downloads]$ sudo cp lnav /usr/bin
[sudo] password for nsandryushin:
[nsandryushin@client.nsandryushin.net Downloads]$ sudo lnav
```

Рис. 2.12: Установка и запуск lnav на клиенте

Убедимся, что на клиенте интерфейс lnav также корректно отображает системные журналы (рис. 2.13).

```
2025-12-13T18:49:09 UTC                               Press F2 to enable mouse support
LOG ▾ : 2025-12-13T18:49:07.000 : syslog_log : messages[9647] : systemd[1] :
Dec 13 18:48:49 client systemd[5028]: Stopped systemd-tmpfiles-clean.timer - Daily Cleanup of User's Te
Dec 13 18:48:49 client systemd[5028]: Closed dbus.socket - D-Bus User Message Bus Socket.
Dec 13 18:48:49 client systemd[5028]: Closed pipewire.socket - PipeWire Multimedia System Sockets.
Dec 13 18:48:49 client systemd[5028]: Stopped systemd-tmpfiles-setup.service - Create User Files and Di
Dec 13 18:48:49 client systemd[5028]: Removed slice app.slice - User Application Slice.
Dec 13 18:48:49 client systemd[5028]: Reached target shutdown.target - Shutdown.
Dec 13 18:48:49 client systemd[5028]: Finished systemd-exit.service - Exit the Session.
Dec 13 18:48:49 client systemd[5028]: Reached target exit.target - Exit the Session.
Dec 13 18:48:50 client systemd-logind[902]: Removed session 11.
Dec 13 18:48:50 client systemd[1]: user@0.service: Deactivated successfully.
Dec 13 18:48:50 client systemd[1]: Stopped user-runtime-dir@0.service - User Manager for UID 0.
Dec 13 18:48:50 client systemd[1]: Stopping user-runtime-dir@0.service - User Runtime Directory /run/us
Dec 13 18:48:50 client systemd[1]: run-user-0.mount: Deactivated successfully.
Dec 13 18:48:50 client systemd[1]: user-runtime-dir@0.service: Deactivated successfully.
Dec 13 18:48:50 client systemd[1]: Stopped user-runtime-dir@0.service - User Runtime Directory /run/use
Dec 13 18:48:50 client systemd[1]: Removed slice user-0.slice - User Slice of UID 0.
Dec 13 18:48:50 client systemd[1]: user-0.slice: Consumed 214ms CPU time, 104.6M memory peak.
Dec 13 18:49:07 client systemd-logind[902]: Existing logind session ID 3 used by new audit session, ign
Dec 13 18:49:07 client systemd-logind[902]: New session c11 of user root.
Dec 13 18:49:07 client systemd[1]: Created slice user-0.slice - User Slice of UID 0.
Dec 13 18:49:07 client systemd[1]: Starting user-runtime-dir@0.service - User Runtime Directory /run/us
Dec 13 18:49:07 client systemd[1]: Finished user-runtime-dir@0.service - User Runtime Directory /run/us
Dec 13 18:49:07 client systemd[1]: Starting user@0.service - User Manager for UID 0...
Dec 13 18:49:07 client systemd-logind[902]: New session 12 of user root.
Dec 13 18:49:07 client systemd[5069]: Queued start job for default target default.target.
Dec 13 18:49:07 client systemd[5069]: Created slice app.slice - User Application Slice.
Dec 13 18:49:07 client systemd[5069]: grub-boot-success.timer - Mark boot as successful after the user
Dec 13 18:49:07 client systemd[5069]: Started systemd-tmpfiles-clean.timer - Daily Cleanup of User's Te
Dec 13 18:49:07 client systemd[5069]: Reached target paths.target - Paths.
Dec 13 18:49:07 client systemd[5069]: Reached target timers.target - Timers.
Dec 13 18:49:07 client systemd[5069]: Starting dbus.socket - D-Bus User Message Bus Socket...
Dec 13 18:49:07 client systemd[5069]: pipewire-pulse.socket - PipeWire PulseAudio was skipped because o
Dec 13 18:49:07 client systemd[5069]: Listening on pipewire.socket - PipeWire Multimedia System Sockets
Dec 13 18:49:07 client systemd[5069]: Starting systemd-tmpfiles-setup.service - Create User Files and D
Dec 13 18:49:07 client systemd[5069]: Finished systemd-tmpfiles-setup.service - Create User Files and D
Dec 13 18:49:07 client systemd[5069]: Listening on dbus.socket - D-Bus User Message Bus Socket.
Dec 13 18:49:07 client systemd[5069]: Reached target sockets.target - Sockets.
Dec 13 18:49:07 client systemd[5069]: Reached target basic.target - Basic System.
Dec 13 18:49:07 client systemd[5069]: Reached target default.target - Main User Target.
Dec 13 18:49:07 client systemd[5069]: Startup finished in 164ms.
Dec 13 18:49:07 client systemd[1]: Started user@0.service - User Manager for UID 0.
Dec 13 18:49:07 client systemd[1]: Started session-c11.scope - Session c11 of User root.
```

Рис. 2.13: Просмотр логов через lnav на клиенте

Приступим к автоматизации настроек. На сервере в каталоге /vagrant/provision создадим структуру папок для хранения конфигов и скопируем туда текущий файл настроек. Также создадим скрипт netlog.sh (рис. 2.14).

```
[nsandryushin@server.nsandryushin.net rsyslog.d]$ cd /vagrant/provision/server
[nsandryushin@server.nsandryushin.net server]$ sudo mkdir -p /vagrant/provision/server/netlog/etc/rsyslog
.d
[sudo] password for nsandryushin:
[nsandryushin@server.nsandryushin.net server]$ sudo cp -R /etc/rsyslog.d/netlog-server.conf /vagrant/prov
ision/server/netlog/etc/rsyslog.d
[nsandryushin@server.nsandryushin.net server]$ cd /vagrant/provision/server
[nsandryushin@server.nsandryushin.net server]$ sudo touch netlog.sh
[nsandryushin@server.nsandryushin.net server]$ sudo nano netlog.sh
[nsandryushin@server.nsandryushin.net server]$ sudo chmod +x netlog.sh
[nsandryushin@server.nsandryushin.net server]$
```

Рис. 2.14: Подготовка файлов для автоматизации сервера

Напишем скрипт netlog.sh, который будет копировать конфигурационные фай-
лы, восстанавливать контекст безопасности SELinux, настраивать фаервол и
перезапускать службу rsyslog (рис. 2.15).

```
GNU nano 8.1 netlog.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/netlog/etc/* /etc
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-port=514/tcp
firewall-cmd --add-port=514/tcp --permanent
echo "Start rsyslog service"
systemctl restart rsyslog
```

Рис. 2.15: Скрипт provisioning для сервера

На клиенте выполним аналогичные действия: создадим структуру каталогов в
/vagrant/provision, скопируем конфиг клиента и создадим установочный скрипт
(рис. 2.16).

```
[nsandryushin@client.nsandryushin.net rsyslog.d]$ cd /vagrant/provision/client
[nsandryushin@client.nsandryushin.net client]$ sudo mkdir -p /vagrant/provision/client/netlog/etc/rsyslog
.d
[sudo] password for nsandryushin:
[nsandryushin@client.nsandryushin.net client]$ sudo cp -R /etc/rsyslog.d/netlog-client.conf
cp: missing destination file operand after '/etc/rsyslog.d/netlog-client.conf'
Try 'cp --help' for more information.
[nsandryushin@client.nsandryushin.net client]$ sudo cp -R /etc/rsyslog.d/netlog-client.conf /vagrant/prov
ision/client/netlog/etc/rsyslog.d/
[nsandryushin@client.nsandryushin.net client]$ cd /vagrant/provision/client
[nsandryushin@client.nsandryushin.net client]$ sudo touch netlog.sh
[nsandryushin@client.nsandryushin.net client]$ sudo chmod +x netlog.sh
[nsandryushin@client.nsandryushin.net client]$ sudo nano netlog.sh
[nsandryushin@client.nsandryushin.net client]$ █
```

Рис. 2.16: Подготовка файлов для автоматизации клиента

В скрипте netlog.sh для клиента пропишем установку пакета lnav, копирование конфигурации rsyslog, восстановление контекста SELinux и перезапуск службы (рис. 2.17).

```
GNU nano 8.1 netlog.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install lnav
echo "Copy configuration files"
cp -R /vagrant/provision/client/netlog/etc/* /etc
restorecon -vR /etc
echo "Start rsyslog service"
systemctl restart rsyslog█
```

Рис. 2.17: Скрипт provisioning для клиента

Наконец, откроем Vagrantfile и добавим в конфигурации сервера и клиента блоки vm.provision типа shell, указывающие на созданные нами скрипты netlog.sh (рис. 2.18).

```

120         preserve_order: true,
121         path: "provision/server/smb.sh"
122     server.vm.provision "server netlog",
123         type: "shell",
124         preserve_order: true,
125         path: "provision/server/netlog.sh"
126     end
127
128     ## Client configuration
129     config.vm.define "client", autostart: false do |client|
130         client.vm.box = "rockylinux10"
131         client.vm.hostname = 'client'
132
133         client.vm.boot_timeout = 1440
134
135         client.ssh.insert_key = false
136         client.ssh.username = 'vagrant'
137         client.ssh.password = 'vagrant'
138
139         client.vm.network :private_network,
140             type: "dhcp",
141             virtualbox__intnet: true
142
143         client.vm.provider :virtualbox do |virtualbox|
144             virtualbox.customize ["modifyvm", :id, "--vrde", "on"]
145             virtualbox.customize ["modifyvm", :id, "--vrdeport", "3392"]
146         end
147
148         client.vm.provision "client dummy",
149             type: "shell",
150             preserve_order: true,
151             path: "provision/client/01-dummy.sh"
152
153         client.vm.provision "client routing",
154             type: "shell",
155             preserve_order: true,
156             run: "always",
157             path: "provision/client/01-routing.sh"
158
159         client.vm.provision "client mail",
160             type: "shell",
161             preserve_order: true,
162             path: "provision/client/mail.sh"
163
164         client.vm.provision "client ntp",
165             type: "shell",
166             preserve_order: true,
167             path: "provision/client/ntp.sh"
168
169         client.vm.provision "client nfs",
170             type: "shell",
171             preserve_order: true,
172             path: "provision/client/nfs.sh"
173
174         client.vm.provision "SMB client",
175             type: "shell",
176             preserve_order: true,
177             path: "provision/client/smb.sh"
178
179         client.vm.provision "client netlog",
180             type: "shell",
181             preserve_order: true,
182             path: "provision/client/netlog.sh"
183     end
184 end

```

Рис. 2.18: Настройка Vagrantfile

3 Выводы

В результате выполнения лабораторной работы были получены навыки использования журналов системных событий